

MATH 301: Homework 2

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Question 1

If $a, b \in \mathbb{R}$, prove the following:

Part (a): If $a + b = 0$, then $b = -a$

Recall Properties of $\mathbb{R}$ Addition:

(Commutativity:) $a + b = b + a$

(Associativity:) $(a + b) + c = a + (b + c)$

(Zero Property:) There exists a $0 \in \mathbb{R}$ such that $a + 0 = a$

(Existence of inverse:) $\forall a \in \mathbb{R}, \exists -a \in \mathbb{R}$ such that $a + (-a) = 0$

Let $a + b = 0$. Observe the following:

$\exists a, b \in \mathbb{R}, b \neq -a \ni$	By proposition
$a + b = 0$	By proposition
$a + b + (-a) = 0 + (-a)$	By existence of inverses
$b + (a + (-a)) = 0 + (-a)$	By associativity
$b + (0) = 0 + (-a)$	By inverse property
$b = -a$	By definition of identity element

Clearly we have shown given $a + b = 0$, then $b = -a$ holds.

Part (c): $(-1)a = -a$

Recall Properties of $\mathbb{R}$ Multiplication

(Commutativity:) $a \cdot b = b \cdot a$

(Associativity:) $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

(Identity:) $\exists 1 \in \mathbb{R}$ such that $a \cdot 1 = a$

(Existence of inverse:) $\forall a \neq 0, \exists \frac{1}{a} \in \mathbb{R}$ such that $a \cdot \frac{1}{a} = 1$

Consider the following:

$a + (-1)a = a + (-1)a$	Proposition $x=x$
$a + (-1)a = 1 \cdot a + (-1)a$	Multiplicative Identity
$a + (-1)a = a(1 + (-1))$	Distributive Property
$a + (-1)a = a(0)$	Inverse Property of Addition
$a + (-1)a = 0$	Zero Multiplicative Property

Thus we can say $a + (-1)a = 0$.

Recall by *Part (a)* "If $a + b = 0$, then $b = -a$ ".

Therefore $(-1)a = -a$ must hold.

Part (d): $-(-a) = a$

Consider the following:

$(-a) + a = 0$	Definition of Additive Inverse
$(-1)((-a) + a) = (-1)0$	Multiply both sides by (-1)
$-(-a) + (-a) = (-1)0$	Distributive Property and Part (b)
$-(-a) + (-a) = 0$	Multiplicative Property of Zero
$-(-a) + (-a) + a = 0 + a$	Add a to both sides
$-(-a) + ((-a) + a) = a$	Commutative Property
$-(-a) + (0) = a$	Inverse Property of Addition
$-(-a) = a$	Conclusion

Thus we have shown that $-(-a) = a$.

Part (e): $(-1)(-1) = 1$

Consider $-(-a) = a$, we stated $a \in \mathbb{R}$ as an arbitrary element.

$$\begin{array}{ll}
 -(-a) = a & \text{Part (d)} \\
 -(-1(a)) = 1(a) & \text{Identity Property of Multiplication} \\
 (-1)(-1(a)) = 1(a) & \text{Part (c)} \\
 (-1)(-1(1)) = 1(1) & \text{Let } a = 1, a \text{ is an arbitrary element of } \mathbb{R} \\
 (-1)(-1) = 1 & \text{Identity Property of Multiplication}
 \end{array}$$

Thus we have shown $(-1)(-1) = 1$

Question 2

If $a, b \in \mathbb{R}$, prove the following:

Part (a): $-(a + b) = (-a) + (-b)$ **Part (a):** $(-a) \cdot (-b) = a \cdot b$ **Part (a):** $\frac{1}{(-a)} = -(\frac{1}{a})$ **Part (a):**

Question 3

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Question 4

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Question 5

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Question 6

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